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JURASSIC AND TRIASSIC PALEOMAGNETISM OF SOUTH KOREA

Kwang Ho Kim

Department of Earth Science Education, Chonbuk National University, Chonju, South Korea

Rob Van der Voo

Department of Geological Sciences, University of Michigan, Ann Arbor

Abstract. Middle to Late Jurassic samples have been collected from granitic rocks in three geological subdivisions of South Korea, from north to south: Kyonggi Massif, Okchon Folded Belt and Ryongnam Massif. Triassic samples (granites and sediments) were collected in the Ryongnam Massif and its border area with the Okchon Folded Belt. The Jurassic declinations of the two massifs are northeasterly (mean D/I = 39.6°/+57.6°, N = 7 sites, k = 31, $\alpha 95$ = 11.1°), whereas the intervening Okchon Folded Belt reveals northwesterly Jurassic declinations (D/I = 337.5°/+47°, N = 7 sites, k = 32, $\alpha 95$ = 10.8°). Reversed-polarity directions are rare, but have been observed in three localities. The widespread occurrence of similar directions in a given province argues against major tectonic tilting of the granites; two sites in Jurassic sediments provide additional structural control. We interpret the direction difference between the Okchon belt and the adjacent massifs as due to left-lateral shear in an actively deforming continental margin during the Late Jurassic-Early Cretaceous Daebo Orogeny. The Triassic directions from the Ryongnam Massif and its border area are east-northeasterly (D/I = 62.9°/+35.7°, N = 5, k = 27, $\alpha 95$ = 15°) with the sediments showing normal and reversed polarities. Comparisons with Jurassic and Triassic paleomagnetic results from adjacent blocks in Asia (Eurasia, North Korea, North and South China) are hampered by a paucity of reliable paleopoles. What seems clear, however, is that North and South China and the major part of Korea yield Jurassic and younger paleopoles that are to first approximation similar; the exception is formed by the results from the Okchon Folded Belt. The aggregate of the China and Korea Blocks yields mean paleopoles for this interval that are different from those published for Eurasia, but this may be due to imperfections in the Eurasian reference poles. For the Triassic, the directions and paleopoles of the China Blocks and the Ryongnam Massif of South Korea are dissimilar, resulting in interpretations of the area in terms of relative motions (mainly rotations).

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INTRODUCTION

Eastern Asia is the ideal natural laboratory to study accretion of crustal blocks, with paleomagnetism one of the principal tools available to quantify relative movements. While the number of paleomagnetic studies is rapidly increasing, there is as yet no abundant, unambig-

uous, Phanerozoic dataset for any of the major blocks such as Siberia, North China or South China [Lin et al., 1985]. Late Mesozoic results are the most abundant, and it is possible to reach some consensus for Jurassic and younger times, although interpretations of the differences in paleopoles between Siberia (i.e., Eurasia) and the China Blocks remain divergent [Lin and Fuller, 1986; Courtillot and Besse, 1986; Lee et al., 1987]. An additional uncertainty in any tectonic analysis of eastern Asia is posed by Korea which traditionally is thought to be part of the North China Block (Sino-Korea). There are many recent publications, however, which separate the North Korea Block (NKB) from the North China Block (NCB) as well as from parts of South Korea [e.g., Lee et al., 1987, and references therein]. The main zone inferred by Lee and co-workers as the boundary between the NCB and the NKB is the Tancheng-Lujiang (or Tan-Lu) fault system (Figure 1); Xu et al. [1987] have argued that the Tan-Lu fault is part of a large left-lateral wrench fault system, extending for 5000 km in length and 800 km in width throughout eastern China, Korea and up to the Sea of Okhotsk.

Major post-Paleozoic orogenic boundaries within Korea are proposed bordering the Okchon Belt (Figure 2). Although it is tempting to correlate one of these boundaries with the suture between the North and South China blocks, there is at this time little evidence for such a correlation. The Okchon Folded Belt (also spelled Ogcheon) separates the Kyonggi Massif to the north from the Ryongnam Massif to the south, as it runs diagonally across the Korean Peninsula (Figure 2). This orogenic belt, deformed in the Late Jurassic to Early Cretaceous Daebo Orogeny, contains questionable ophiolites of pre-Jurassic(?) age, which has led some authors to suggest that it is a pre-Cretaceous suture [e.g., Lee, 1987, p. 225].

In order to test some of these ideas about Korean geologic subdivisions, as well as the paleogeographic setting of Korea in comparison with the China Blocks and Eurasia, Jurassic and Triassic granites and sediments were sampled for a paleomagnetic study.

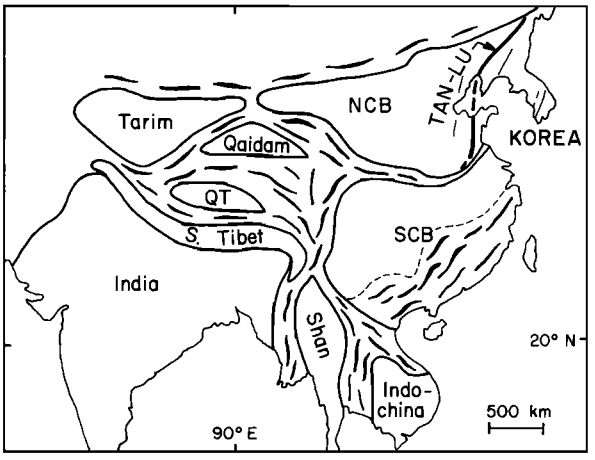


Fig. 1. Sketch map of the tectonic elements of southeastern Asia, modified from Li et al. [1982]. NCB is the North China Block and SCB is the South China Block; QT is the Qiangtang Block. The location of the Tan-Lu fault [Xu et al., 1987] is indicated and may form a major boundary between the North China Block and the North Korean Block.

GEOLOGIC SETTING AND SAMPLING

From north to south, South Korea is divided geologically into the four provinces of the Kyonggi Massif, the Okchon Folded Belt, the Ryongnam Massif and the southeasterly Kyongsang Basin, also spelled as Gyeongsang (Figure 2).

The Kyonggi Massif is separated from the North Korean Pyongnam Basin by the Chugaryong (also Shugareong) Zone or suture [e.g., Lee et al., 1987], although this lineament is not mentioned as a suture by Lee [1987] and at any rate is likely to be of Paleozoic age [Lee et al., 1987]. For the purposes of this paper, it is convenient to consider the Kyonggi Massif during Mesozoic time as a part of the same geological province as North Korea. The oldest basement rocks of the Massif are Archean metamorphic rocks that are unconformably overlain by Proterozoic strata. The area was a stable Paleozoic platform until the Triassic Songnim Orogeny and the subsequent Late Jurassic to Early Cretaceous Daebo Orogeny. The latter orogeny was accompanied by the intrusion of the synorogenic Daebo granites, which we have collected for our study at three localities (Figure 2: Jp - 2 sites, Jb and Jm). One site (Jmj) in Jurassic clastic sediments of the Chungnam System has also been collected in order to provide structural control on our paleomagnetic directions. Previously published paleomagnetic results are available for Cretaceous rocks of the Kyonggi Massif from the Chongok

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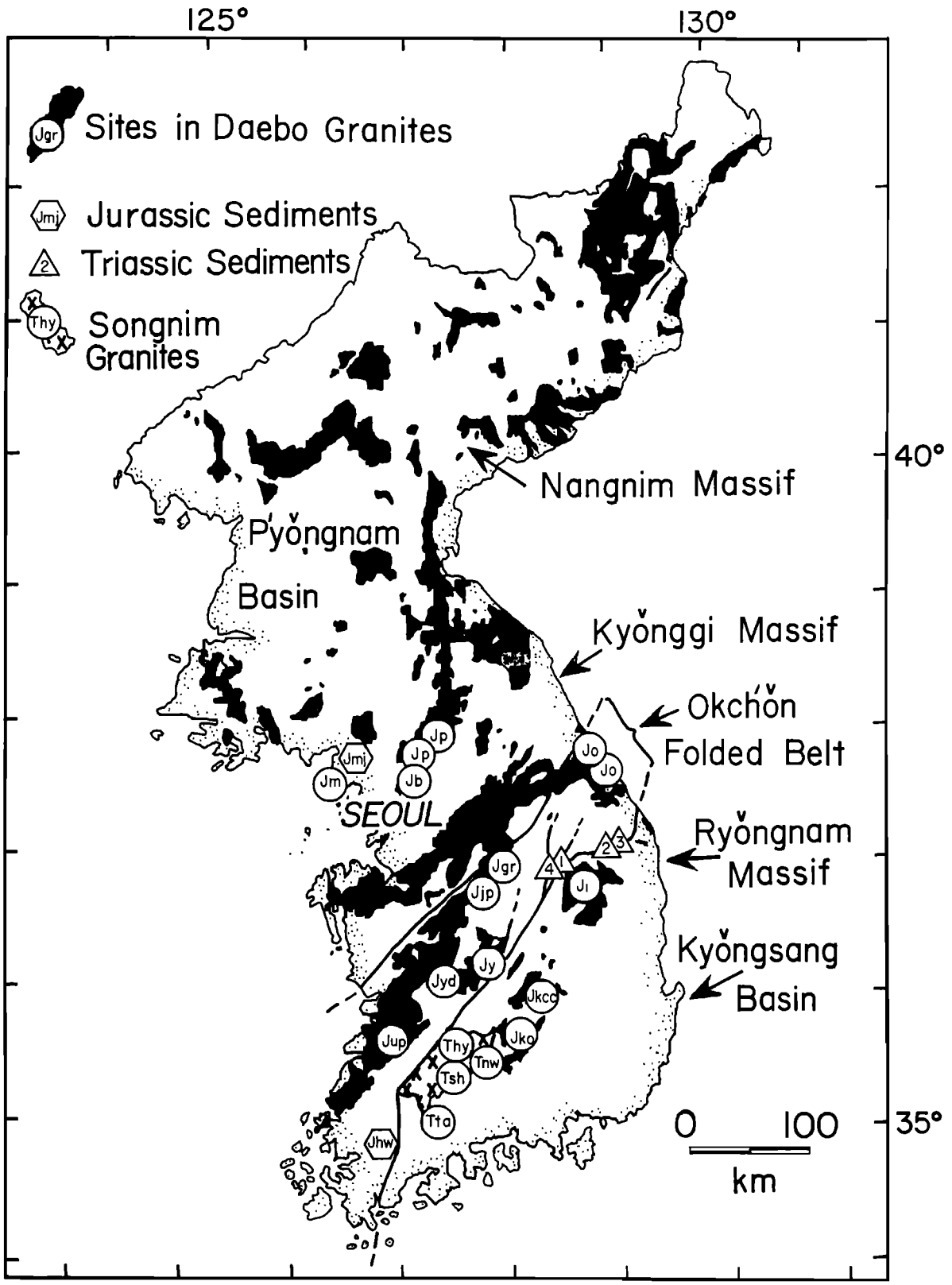


Fig. 2. Map showing sampling localities, tectonic provinces of Korea, and the distribution of the Jurassic Daebo Granites (black) and the Late Triassic Songnim Granites (crosses).

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(also Jeongog) area northeast of Seoul, and for Jurassic and Cretaceous rocks from North Korea [Gurarii et al., 1966; Kang, 1966, 1972, 1975; Li, 1975a,b; and Kang and Li, 1977].

The Okchon Folded Belt extends in a SW-NE ("Sinian") direction diagonally across the peninsula, with a width of about 70 km. The belt is divided into two parts, the southwesterly metamorphosed zone and the northeasterly non-metamorphic, or sedimentary zone. The former is composed of metamorphic Paleozoic and older strata, intruded by Jurassic Daebo granites, which were sampled for paleomagnetism at five localities (Figure 2: Jjp, Jgr, Jy, Jyd and Jup). In the southwestern part of the metamorphic Okchon Belt, one site of unmetamorphosed Jurassic sediments (Jhw) was sampled in tuffaceous sandstones of the Whasun Coal Mine area. The metamorphosed zone has clearly defined boundaries, often faults, with the Kyonggi and Ryongnam Massifs as well as the non-metamorphosed zone. The above-mentioned ophiolitic (?) rocks occur along the southern boundary with the Ryongnam Massif.

In contrast, the northeasterly non-metamorphosed zone of the Okchon Folded Belt has less well-defined boundaries. It consists of the Paleozoic to Early Mesozoic Choson and Pyongan Supergroups. Fossils found in the Paleozoic sediments have been correlated with those from the North Korean Pyongnam Basin, resulting in the interpretation [Lee, 1987] that in the Paleozoic the non-metamorphosed zone of the Okchon Folded Belt, North Korea, and North China were part of the same block. The metamorphosed zone, on the other hand, contains distinct faunas and hence was a separate block possibly connected to the South China Block or Yangtze Paraplatform [Lee, 1987, p. 249]. We sampled Triassic red beds (sites (T) 1 - 4) in the Pyongan Supergroup of the non-metamorphosed zone for paleomagnetic study (Figure 2). Paleomagnetic results published previously are available for the same Triassic units as well as some of the older rocks from this zone [Shibuya et al., 1988]. In this area structures show east-west trending folds, following the "Songnim" direction related to the Late Triassic orogeny of the same name. In the extreme northeast of the non-metamorphosed zone, a Daebo granite was also sampled (Figure 2: Jo - 2 sites).

The Ryongnam Massif is fault-bounded with the metamorphosed zone of the Okchon Belt. The Massif is composed of Precambrian gneisses, intruded by Jurassic Daebo and Triassic Songnim granites [Choo, 1986; Choo and Kim, 1986]. Samples for paleomagnetic study were collected at three and four localities, respectively (Figure 2: Ji, Jkcc, Jko and Tsh, Tnw, Tta and Thy).

The Kyongsang Basin occupies the southeasternmost part of Korea and was not sampled by us. Previously published paleomagnetic results are available for Cretaceous rocks [Min et al., 1982; Otofuji et al., 1982; Kim, 1985; Lee et al., 1987]. To summarize our sampling, we report in this paper on (1) 12 localities in Daebo Granites, which are distributed over the Kyonggi, Okchon and Ryongnam provinces and generally range in age from 148 to 178 Ma (Table 1) [Kim, 1971; Park, 1972; Hurley et al., 1973; Choo, 1971, 1983; Choo et al., 1982; Yun and Silberman, 1979; Kim and Lee, 1983]; (2) one site in Middle Jurassic sediments of the Munjusan Formation (Chungnam System) of the Kyonggi Massif; (3) one site in Jurassic sediments in the Whasun Coal Mine in the southern Okchon Belt; (4) four sites of Late Triassic Songnim granites in the Ryongnam Massif, with radiometric ages ranging from 211 to 228 Ma (Table 1) [Kim and Kim, 1988; Choo, 1986]; and (5) four sites of Triassic red beds in the area of the Okchon (non-metamorphosed) zone just north of the Ryongnam Massif. These gently NW-dipping strata are dated no more precisely than simply Triassic {Lee, 1987], and have been called the Nogam Series or the Tonggo Formation.

PALEOMAGNETIC RESULTS

The samples have been measured with a ScT cryogenic magnetometer at the University of Michigan and a Molspin spinner magnetometer at Chonbuk National University. Thermal and alternating field (AF) demagnetizations were carried out in a magnetically shielded room with restfield of 200 nT or less. All samples were progressively demagnetized and results have been plotted in orthogonal demagnetization diagrams [Zijderveld, 1967] and in stereonets, followed by Principal Component Analysis [Kirschvink, 1980]. Several demagnetization trajectories in stereonets have been analyzed by the great-circle technique of Halls [1976]. For the Triassic sediments, these have been combined with the directions determined from orthogonal demagnetization diagrams following the method of Bailey and Halls [1984], whereas

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TABLE 1. Site-Mean Paleomagnetic Results

Site	n	D/I	k	α95	VGP	Ages, Comments
					Jurassic Kyonggi Massif	
Jp	9	61.5/+46.4	391	2.6	38.2°N, 209.2°E	157K(47); 165R(48); One reversed direction
Jb	6	208.6/-65.7	12	20.3	66.7°N, 181.3°E	202R(50); 65K(52-7); One normal direction
Jm	12	46.2/+45.5	219	2.9	49.9°N, 217.5°E	Late Jurassic, by association
Jmj	4	33.0/+39.3	102	9.2	57.8°N, 235.6°E	Middle Jurassic sediments; D/I without TC is 18.9/+34.0
Mean	4	43.5/+49.9	30	17.0		
					Jurassic Ryongnam Massif	
Ji	9	25.8/+61.5	155	4.2	69.4°N, 194.4°E	133K(235); 163K(231); 167K(228)
Jkcc	6	31.8/+70.7	80	7.5	61.1°N, 166.9°E	166K(109); 178K(120)
Jko	7	38.1/+69.6	112	5.7	58.0°N, 172.2°E	Late Jurassic, by association
Mean	3	31.2/+67.4	211	8.5		
Mean Kyonggi and Ryongnam Massifs						
	7	39.6/+57.6	31	11.1		
					Jurassic Okchon Folded Belt	
Jo	14	333.6/+57.3	91	4.2	69.2°N, 46.9°E	148K(123); 156K(285); 2 sites combined
Jjp	3	337.9/+49.8	29	23.3	70.7°N, 26.1°E	159K(69); 163K(88)
Jgr	5	166.7/-34.5	48	11.2	68.7°N, 344.5°E	121K(68); All reversed directions
Jy	17	325.0/+41.6	65	4.5	57.6°N, 25.4°E	190±1R(91); 119±8K(91)
Jyd	3	329.2/+64.3	23	26.1	64.8°N, 70.6°E	163±1R(94); 170±5R(92); 175±2R(93)
Jup	3	331.1/+50.7	208	8.6	65.6°N, 35.5°E	170K(107); 148K(113)
Jhw	3	350.4/+28.4	121	11.3	68.5°N, 332.7°E	Jurassic sediments; D/I without TC is 327.4/+52.5
Mean	7	337.5/+47.0	32	10.8		
					Triassic Ryongnam Massif and Adjacent Okchon Belt	
Tnw	5	70.6/+37.1	38	12.6	27.3°N, 210.7°E	228±5R(134); 222±5R(135); 211±3R(133)
Tsh	8	50.9/+39.4	15	14.7	44.1°N, 219.3°E	as above
Tta	9	36.6/+39.0	30	9.5	55.6°N, 229.2°E	as above
Thy	7	77.3/+33.6	20	13.7	20.7°N, 208.9°E	as above
Tl-4	12	75.1/+24.1	27	12.1	19.4°N, 217.1°E	D/I without TC is 71.5/+2.4; four sites of sediments; includes 8 great circles
Mean	5	62.9/+35.7	27	15.0		

Explanation: Sites correspond to labels of Figure 2. n is the number of samples; D/I is the declination and inclination in degrees, k and α95 are the statistical parameters associated with the site-mean [Fisher, 1953]; VGP is the Virtual Geomagnetic Pole calculated from the site-mean direction; Radiometric ages are given in Ma; R, Rb/Sr method, generally whole-rock; K, K/Ar method, generally on biotite; numbers in parentheses correspond to the reference numbers of Lee [1987; his Appendix III and his Isotope Age Map of Korea]; TC is the correction for the tilt of the strata for sedimentary sites Jmj, Jhw and the four combined Triassic sites (Tl-4). For the mean of the Triassic sediments (sites Tl-4), we have combined stable endpoint directions with poles to great-circles following the method of Bailey and Halls [1984].

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for the Jurassic granites we have sufficient numbers of stable-endpoint directions, so that the great-circle data are given in this paper for information only. Almost all samples revealed a present-day field magnetic component. In some samples, this was the only component that could be determined; these samples have not been included in our results. However, many samples showed that a second magnetization was present, which has been summarized in Table 1 and will be discussed below.

Kyonggi and Ryongnam Massifs - Jurassic

Two localities in the Daebo granites (Jm, Jp) and one site of sediments (Jmj) in the Kyonggi Massif revealed characteristic directions which are northeasterly and intermediate to steeply down (Figure 3). In many samples, coercivities and blocking temperatures up to 580°C suggest that magnetite is the carrier of the remanence, but others clearly indicate hematite as well with blocking temperatures up to 670°C (e.g., sample Jm 2-1f in Figure 3). A third site in the Daebo granites (Jb) revealed south-westerly and steeply upward directions (Figure 4). A comparison with directions from the other sites in the Kyonggi and Ryongnam Massifs (Figure 5) shows that these directions from site Jb are roughly antipodal to the others and can be considered to be of reversed polarity. The anomalously young K-Ar age of 65 Ma for this site (Table 1) was obtained from a K-feldspar and is probably not representative of the age of the rock, which is contiguous with the granites of the Jp sites.

The one site in Middle Jurassic sediments has a tilt correction for a southeasterly dip, which brings the directions (Figure 4) into good agreement with the directions from the granites (Figure 5).

Three localities of Daebo Granites in the Ryongnam Massif (Figure 2) yield the same northeasterly characteristic directions as those of the Kyonggi Massif, all with normal polarity directions (Figure 5). Again the remanence appears to be carried by both magnetite and hematite as can be deduced from the coercivities and blocking temperatures revealed during AF and thermal demagnetization of typical samples (Figure 6).

Okchon Folded Belt - Jurassic

Five localities of Daebo Granites reveal characteristic directions, carried by magnetite and hematite. These localities display northwesterly declinations as can be seen in the orthogonal demagnetization diagrams (Figure 7). The inclinations are the same as those of the adjacent Kyonggi

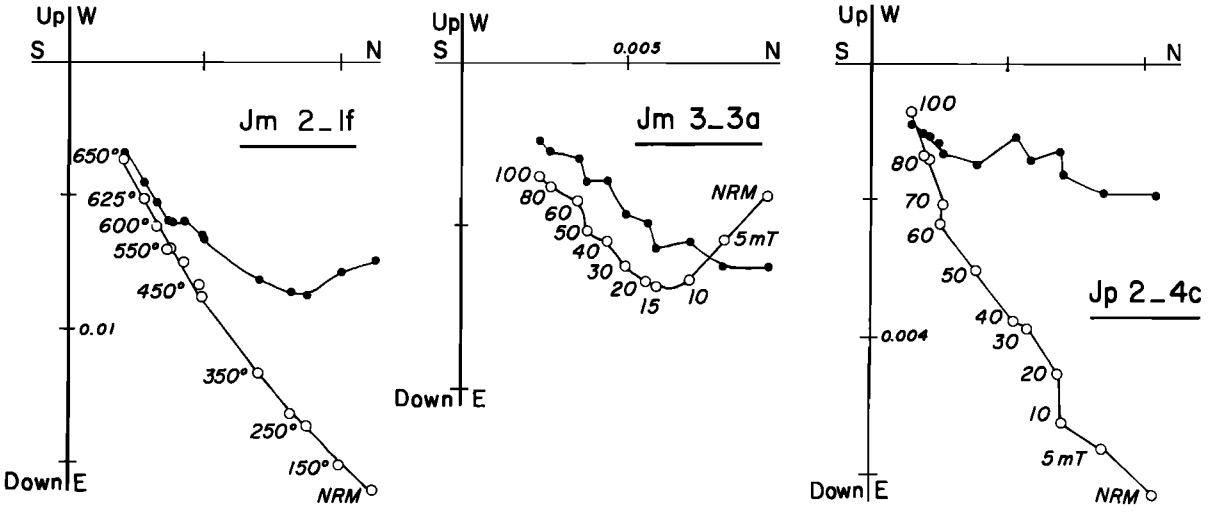


Fig. 3. Vector demagnetization diagrams [Zijderveld, 1967] of Jurassic Daebo granite samples from the Kyonggi Massif. Open (solid) symbols represent projections onto the vertical (horizontal) plane; temperatures (in degrees Celsius) and alternating field values (in milliteslas) are indicated. Intensities of the magnetization along the axes are given in amps per meter.

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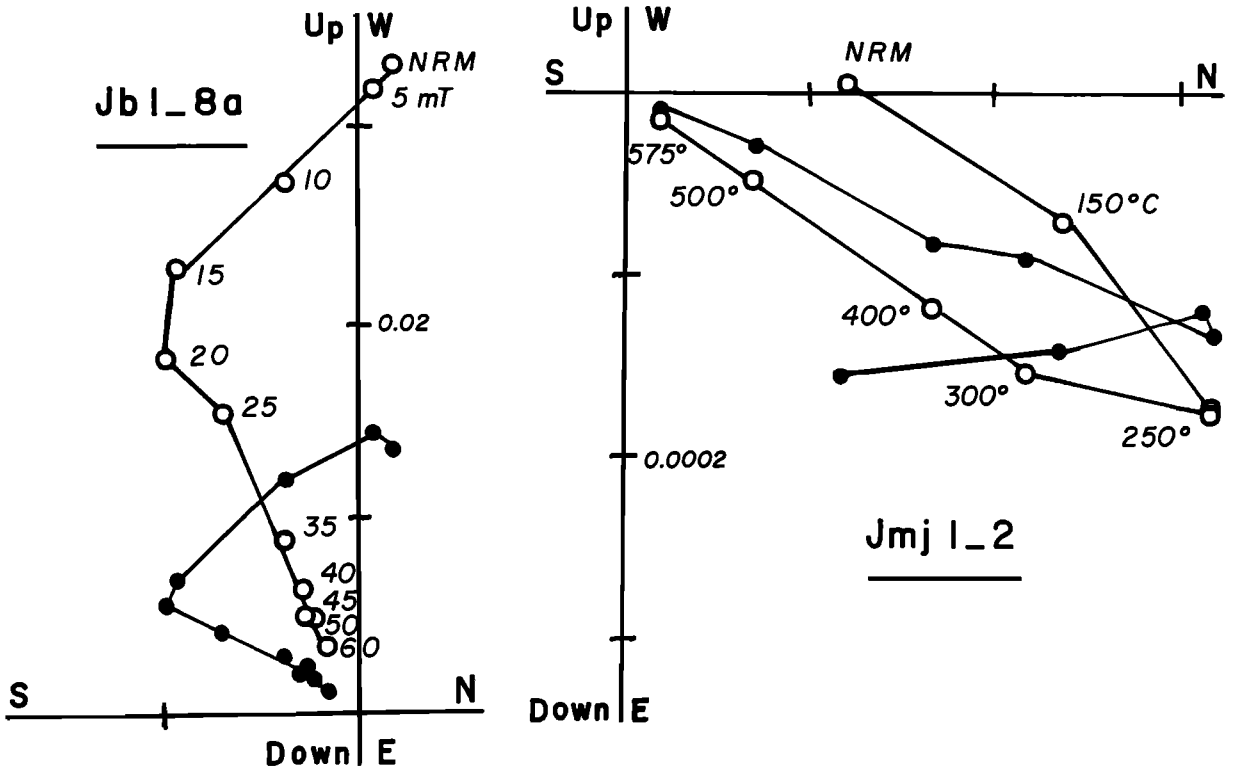


Fig. 4. Vector demagnetization diagrams [Zijderveld, 1967] for the Kyonggi Massif of a Jurassic Daebo granite sample with reversed polarity from site Jb and of a Middle Jurassic sedimentary sample from site Jmj (in situ coordinates). For explanation of conventions see Figure 3.

and Ryongnam Massifs, but apparently the Okchon Belt had a different meridional orientation at the time of the magnetization acquisition by the Daebo Granites. One additional site (Jgr) shows reversed polarities (Figure 8), with a SSE nearly antipodal direction. One site of Jurassic sediments (Jhw) shows the same declination as the granites (before as well as after tilt correction, Table 1), providing structural control to the northwesterly Jurassic directions of the Okchon Folded Belt. Some samples from localities Jo and Jy show curved demagnetization trajectories indicating overlapping coercivities or blocking temperatures. Great-circle analysis reveals poles normal to the demagnetization trajectory planes (Figure 9), which themselves are distributed in a great-circle pattern. The direction in common is the normal to the latter great-circle (Figure 9), which agrees well with all other characteristic directions observed in the Jurassic granites from the Okchon Folded Belt. Because we have sufficient numbers of stable-endpoint directions for localities Jo and Jy, we have not included the additional information from the great circles in the site-mean directions; in view of the considerable scatter of great-circle poles about the best-fit plane (Figure 9), the addition of these extra points does not make a significant contribution to the site means.

Ryongnam Massif and Adjacent Okchon Belt - Triassic

Four localities in Late Triassic Songnim Granites and four sites (T 1 - 4) in Triassic redbeds (Figure 2; Table 1) yielded moderately coherent results with dual-polarity ENE-WSW directions. Demagnetization diagrams are shown in Figures 10 and 11. The granites reveal magnetizations carried by both magnetite and hematite as

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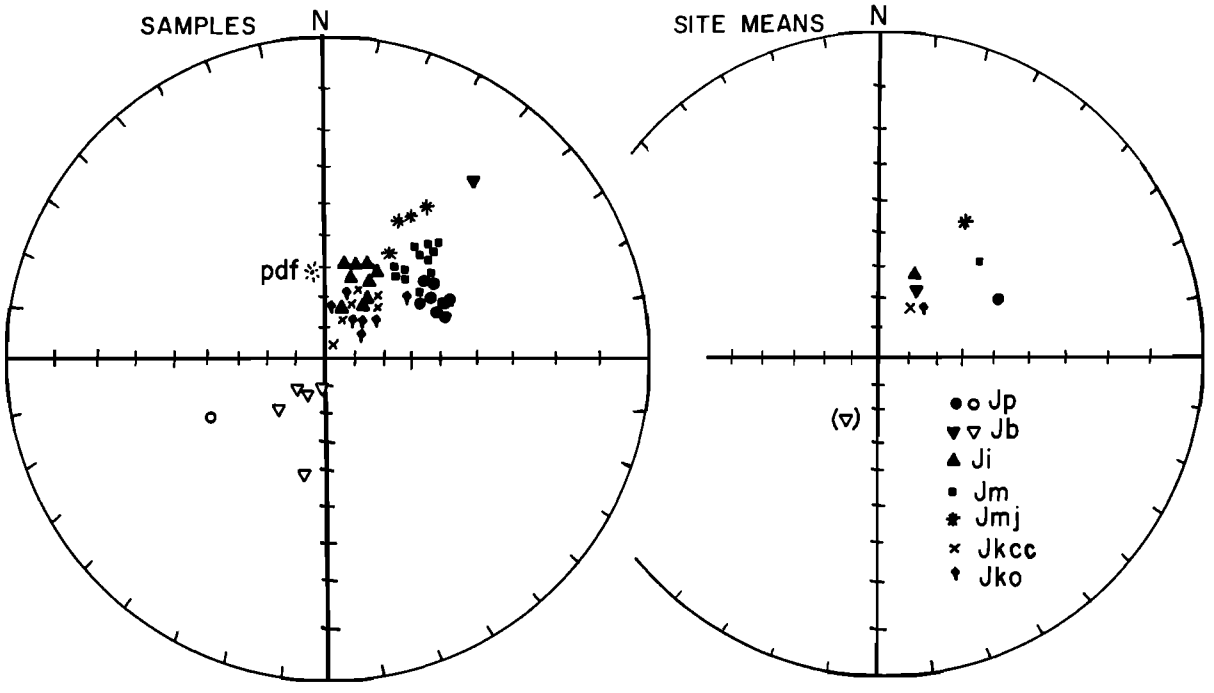


Fig. 5. Equal-angle steroplot with characteristic directions for the samples and site-means of the Jurassic localities in the Kyonggi and Ryongnam Massifs. Open (closed) symbols represent projections onto the upper (lower) hemisphere. The open, inverted triangle in parentheses (site Jb) has been inverted (solid inverted triangle) so that it can be compared with the other site means. The present-day geomagnetic field direction is represented by pdf.

deduced from the coercivities and blocking temperatures, whereas the magnetizations in the red beds are carried by hematite (Figure 11). In many cases the red bed samples acquired strong viscous magnetizations after treatments above 660°C, the onset of which can be seen in Figure 11 by the dashed-line trajectories. Because it is uncertain that the characteristic component has been isolated by these temperatures, we have analyzed the corresponding great-circles on a stereonet (Figure 12). Poles to great circles (triangles) and stable-endpoint directions (circles), when combined, point to a common mean characteristic direction for the red beds with D/I = 75°/+24° after tilt-correction using the method of Bailey and Halls [1984]. There is no significant improvement in clustering upon tilt correction because of the similarities in the shallow dips. However, the Songnim Granites also have (east-)northeasterly directions, with declinations ranging from 37 to 77°, and a mean inclination of 37° (Table 1). When their directions are combined with those of the Triassic red beds, they are seen to be significantly different (more easterly and shallower) from those of the Jurassic Daebo Granites.

DISCUSSION OF RESULTS

We have obtained three groups of directions: (1) the northeasterly ones in the Jurassic Daebo Granites and sediments of the Kyonggi and Ryongnam Massifs, (2) the northwesterly directions in the Jurassic Daebo Granites and sediments from the Okchon Folded Belt, and (3) east-northeasterly shallower directions from Triassic Songnim Granites and red beds of the Nogam Series in the Ryongnam Massif and the adjacent non-metamorphosed zone of the Okchon Folded Belt. Although structural control is lacking for the granites, the similar directions from the Jurassic and Triassic sediments provide a positive test of the assumption that no appreciable tilt correction is necessary for the granites;

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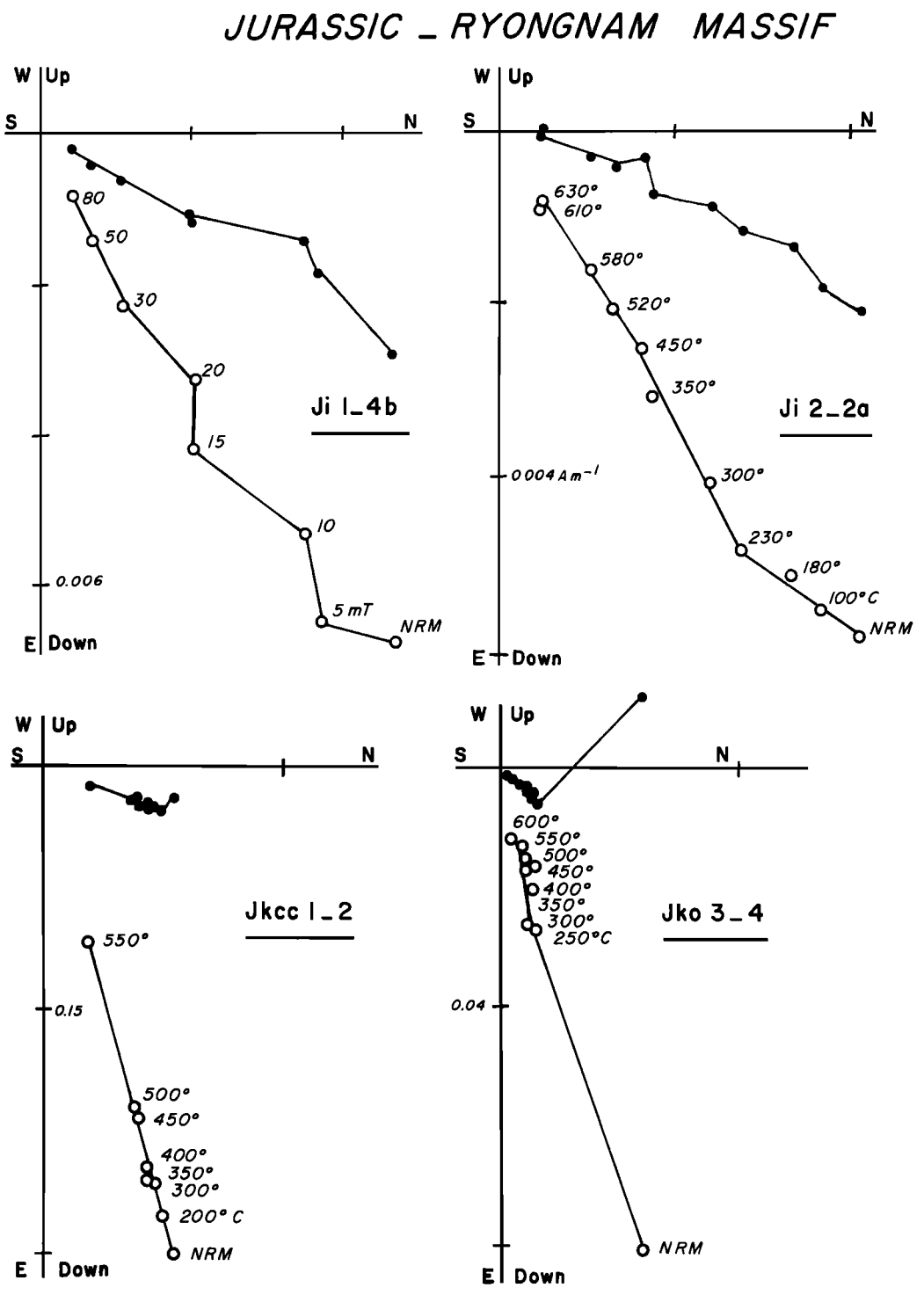


Fig. 6. Vector demagnetization diagrams [Zijderveld, 1967] of Jurassic Daebo granite samples from the Ryongnam Massif. For explanation see Figure 3.

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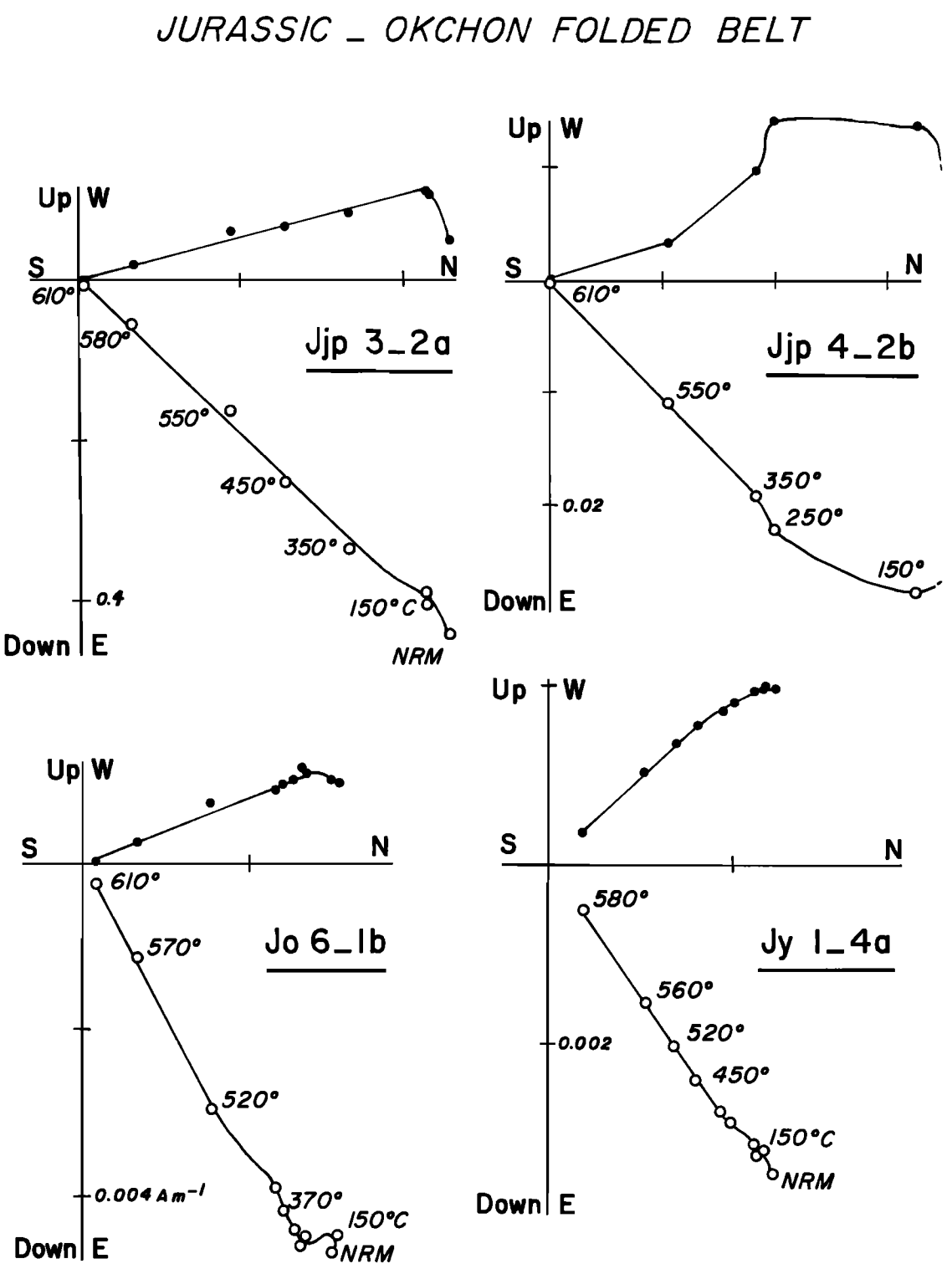


Fig. 7. Vector demagnetization diagrams [Zijderveld, 1967] of Jurassic Daebo granite samples from the Okchon Folded Belt. For explanation see Figure 3.

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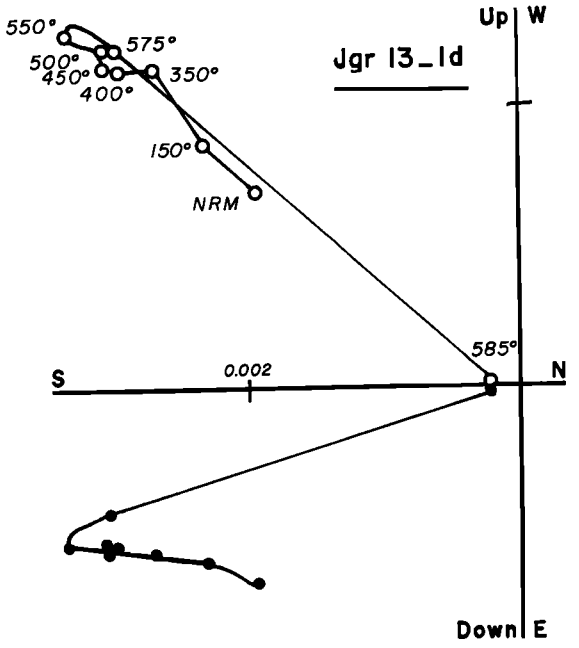


Fig. 8. Vector demagnetization diagram [Zijderveld, 1967] of a Jurassic Daebo granite sample with reversed polarity from the Okchon Folded Belt. For explanation see Figure 3.

moreover, the directions are similar in sites spread over a wide area in a given province, suggesting that the assumption is correct.

Some of the granites and Triassic red beds yielded dual-polarities, indicating that sufficient time has lapsed during the acquisition of the magnetization for the field to have reversed itself. While this suggests, but does not prove, that the magnetizations are primary, it also indicates (1) that secular variation is likely to have been averaged out, and (2) that no major unrecognized magnetic overprints contaminate our mean directions (Table 1). The difference in direction between the Jurassic (and younger) results from Korea and the directions observed in our Triassic rocks suggests that the Triassic magnetizations are likely to be primary.

The difference in declination between the Okchon Belt and the adjacent Massifs, with the inclinations roughly similar, indicates that the Okchon Belt underwent about 60° of counterclockwise rotation. This rotation happened after intrusion of the granites, during or after the latest Jurassic to Cretaceous Daebo orogeny.

We note that the northwesterly Jurassic

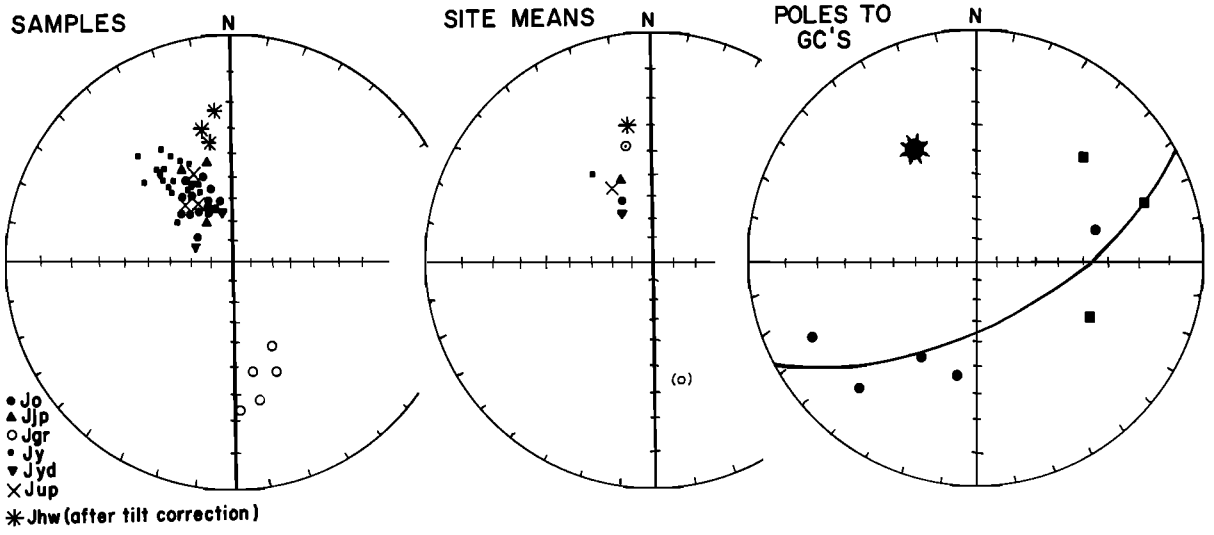


Fig. 9. Equal-angle stereoplot with characteristic directions for the samples and site-means of the Jurassic localities in the Okchon Folded Belt. For explanation see Figure 5. The diagram on the far right displays poles to great circles obtained from curved demagnetization trajectories for 5 samples from site Jo and 3 samples from site Jy. These poles lie themselves in a great-circle distribution, the normal to which ("sun") is the characteristic direction in common; this direction is similar to those for the other two diagrams.

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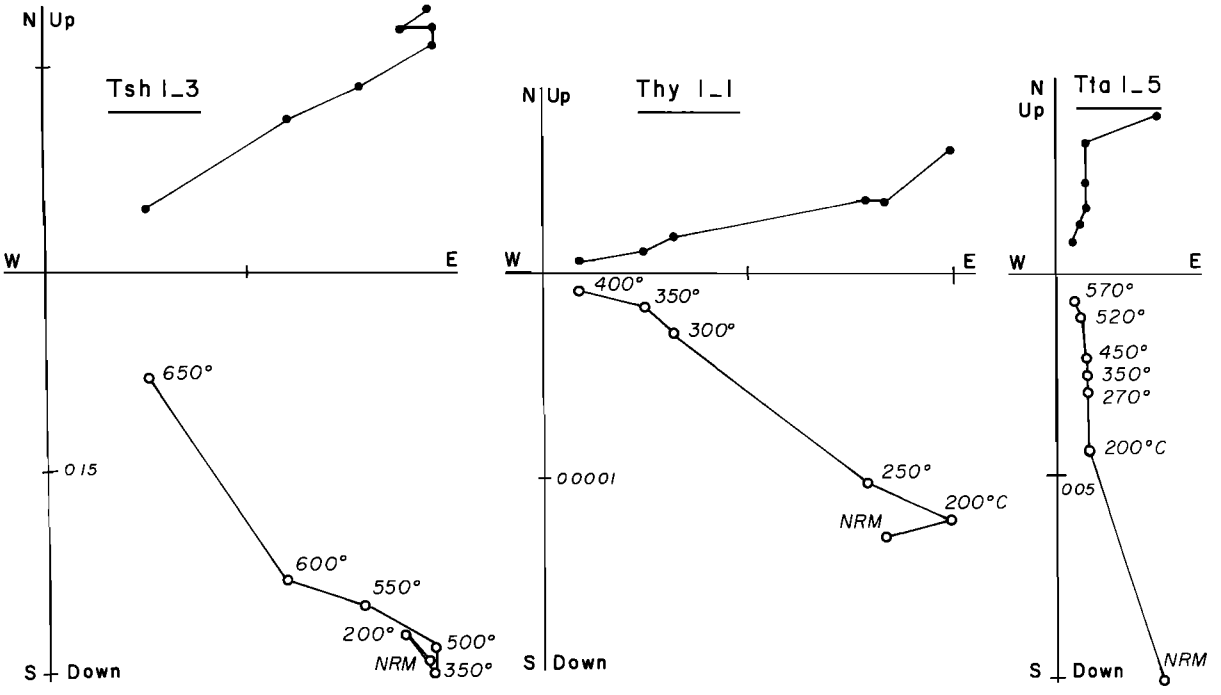


Fig. 10. Vector demagnetization diagrams [Zijderveld, 1967] of Triassic Songnim granite samples from the Ryongnam Massif. For explanation for Figure 3.

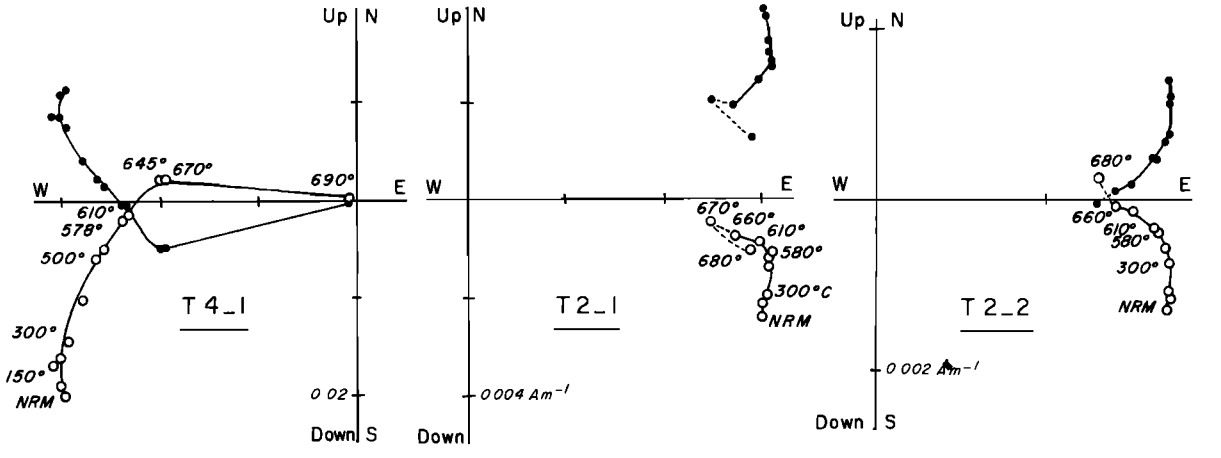


Fig. 11. Vector demagnetization diagrams [Zijderveld, 1967] of Triassic red bed samples from the area in the Okchon Folded Belt bordering the Ryongnam Massif. For explanation for Figure 3. The dashed trajectories represent the onset of unstable magnetic behavior at high-temperature treatment due to mineralogical changes in the samples.

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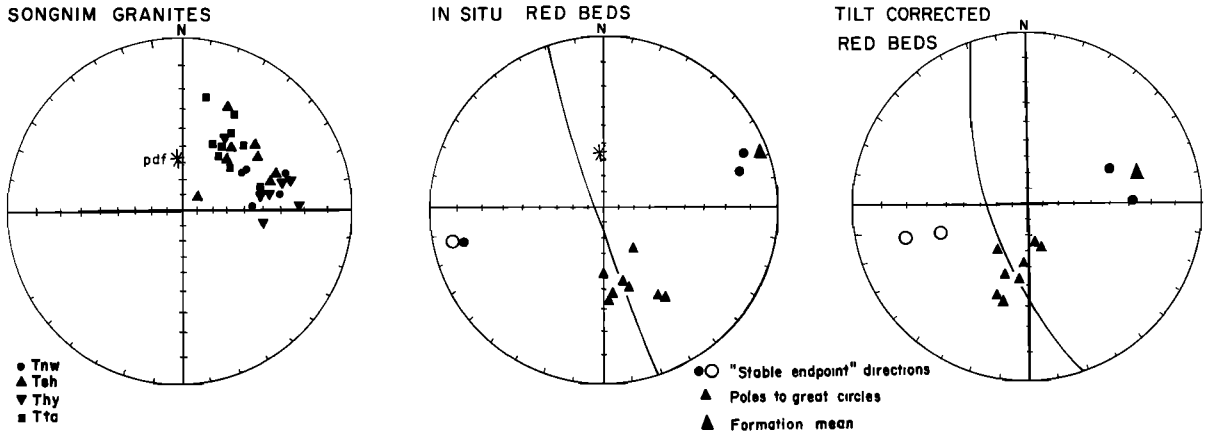


Fig. 12. Equal-angle stereoplot with characteristic directions for the samples of the Triassic localities in the Ryongnam Massif and adjacent Okchon Belt. For explanation see Figure 5. Triangles represent poles to great circles obtained from curved demagnetization trajectories for 8 red bed samples. These poles have been combined with the stable-endpoint directions to yield a common mean direction (large triangle) following Bailey and Halls [1984].

directions from the Okchon Folded Belt are derived from sites in the metamorphosed zone of the Belt as well as from the Jo locality in the extreme northeastern part of the non-metamorphosed zone. This illustrates that the inferred rotations are represented widely throughout the Belt, as illustrated in Figure 13. The three Jurassic mean directions from the Ryongnam Massif (Figure 13), which are not rotated, mark the southern limit of the zone of rotations. However, the Triassic directions from the Ryongnam Songnim Granites, which presumably are also from the non-rotated area, roughly agree with those from the Triassic red beds; the latter are located on the southern margin of the non-metamorphosed zone of the Okchon Folded Belt (triangle in Figure 13). It appears, then, that the red bed sites are from a border area inside the Okchon Belt which is not demonstrably rotated. We deduce that the southern limit of the Okchon rotated terrane is clearly outlined for the metamorphosed zone, but that it is ambiguous for the non-metamorphosed zone. The northern limit of the rotated terrane could be the boundary between the Okchon Folded Belt and the Kyonggi Massif, or it could be somewhat farther north in view of the fact that no paleomagnetic results are available for the southern part of the Kyonggi Massif (Figure 13). We interpret the rotation of the Okchon Folded Belt as due to left-lateral shear in an actively deforming continental margin possibly as early as during the Late Jurassic to Early Cretaceous Daebo Orogeny.

Virtual geomagnetic poles (VGP's) have been calculated for each locality from the site mean directions (Table 1), and the VGP's have been averaged to give paleopoles and associated statistical parameters (Table 2). Previously published Jurassic paleomagnetic results are available from North Korea and have also been listed in Table 2. The latter results are from sediments, extrusives and intrusives, with positive fold and reversal tests. These North Korean results agree roughly with those from the South Korean Massifs.

In Figure 14 the Triassic and Jurassic paleopoles for Korea can be compared with those for Siberia (Eurasia), North China (NCB) and South China (SCB). In order to illustrate all paleopoles in one hemisphere, we show them as calculated from normal directions, but have reversed the Triassic paleopole from the North China Block. The paragraphs below discuss the tectonic conclusions which can be drawn from these paleopoles.

The Jurassic paleopoles from the non-rotated Korean areas (Figure 14a) are roughly in the same area as the paleopoles from both China Blocks. Minor differences

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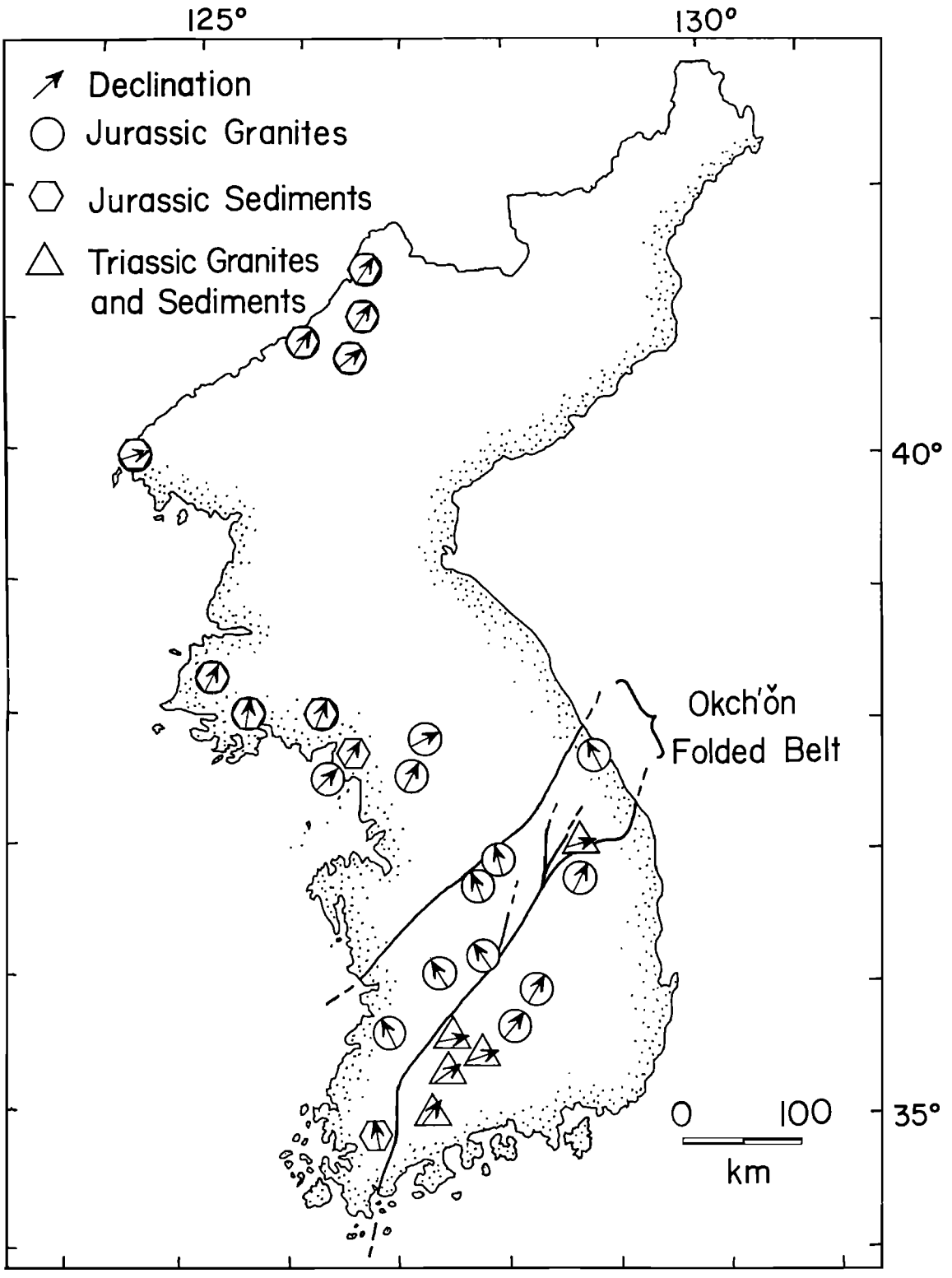


Fig. 13. Map of Korea, as in Figure 2, with Jurassic (circles) and Triassic (triangles) declinations represented by arrows, to illustrate the relative rotation between the Okchon Folded Belt and the adjacent Massifs. North Korean data are from Kang and Li [1977], as listed in Lee [1987] and summarized in Table 2.

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TABLE 2. Paleopoles for Korea, Used in this Paper

Age and Location	Nr.	Mean Pole	N	K	A95	Reference
		Jurassic, South Korea				
Kyonggi Massif	1	54.4°N, 212.9°E	4	24	19.4	This study (Table 1)
Ryongnam Massif	2	63.3°N, 176.0°E	3	91	13.0	This study (Table 1)
Massifs Combined		59.5°N, 199.3°E	7	23	12.8	This study (Table 1)
Okchon Folded Belt	3	69.4°N, 24.9°E	7	37	10.1	This study (Table 1)
		Triassic, South Korea				
Ryongnam Granites		37.1°N, 215.5°E	4	22	20.0	This study (Table 1)
Okchon Belt, Tilt-corrected		19.4°N, 217.1°E	1	-	-	This study (Table 1)
Ryongnam+ Okchon Combined	4	33.5°N, 215.9°E	5	23	16.3	This study (Table 1)
Okchon Belt, in situ		9.8°S, 176.9°E	3	20	28.6	Shibuya et al. [1988]
Okchon Belt, Tilt-corrected	5	13.1°N, 184.5°E	3	19	29.4	Shibuya et al. [1988]
		Jurassic, North Korea				
Nangnim M. and Pyongnam Basin	6	63.2°N, 173.9°E	8	41	8.8	Kang and Li [1977]*

Nr. refers to the numbers denoting paleopoles in Figure 14. The mean pole is calculated by averaging the site-mean virtual geomagnetic poles of Table 1, and the associated statistical parameters (K, A95) are given according to Fisher [1953].
* Mean of eight separate localities with northeasterly declinations (excluding aberrant sites Shinuiju and Pakchon) from Kang and Li [1977] (also listed in Lee [1987, p. 283]).

exist for the Jurassic, but it is not clear that these have statistical and/or tectonic significance, given the possibility of minor apparent polar wander during the Jurassic. We therefore conclude that the Korean Massifs and the North China and South China Blocks were approximately in the same relative position as today since the Late Jurassic. The aggregate group of these Jurassic paleopoles, however, is different from the Eurasia Jurassic paleopoles. This is also true for the Cretaceous. For the Cretaceous, a discussion of this difference has centered on the question whether the Eurasian Cretaceous poles are sufficiently well determined. Courtillot and Besse [1986] and Lee et al. [1987] have argued that the Eurasian Cretaceous pole is in need of revision and that there is no need to consider relative motions between China and Siberia. In contrast, Lin and Fuller [1986] have maintained that a tectonic conclusion is warranted in terms of relative motions between Siberia and the China Blocks. It appears that with our paleopoles we could extend this discussion to earlier times, but for the Jurassic the Eurasian paleopole is probably not that well determined either, so it is questionable whether a difference between the Eurasian Jurassic paleopoles and those from China and Korea is significant in tectonic terms. The locations of the North American Middle to Late Jurassic poles, when rotated to European coordinates by closing the Atlantic Ocean [Bullard et al., 1965], as well as the Eurasian poles themselves are generally (but not all) more westerly (Figure 14b) than those from China and Korea [Frei and Cox, 1987, their Figure 2 and Table 2]. We find the points made by Courtillot and Besse [1986] for the Cretaceous fairly convincing and it appears at this time that we should invoke the same reasoning for the Middle to Late Jurassic; thus, we argue that Late Jurassic to Cretaceous movements between Siberia and the China-Korea blocks may have occurred but are not unambiguously demonstrated by the paleomagnetic data.

A comparison between the non-rotated areas of Korea and the NCB shows that while the Jurassic poles are only slightly

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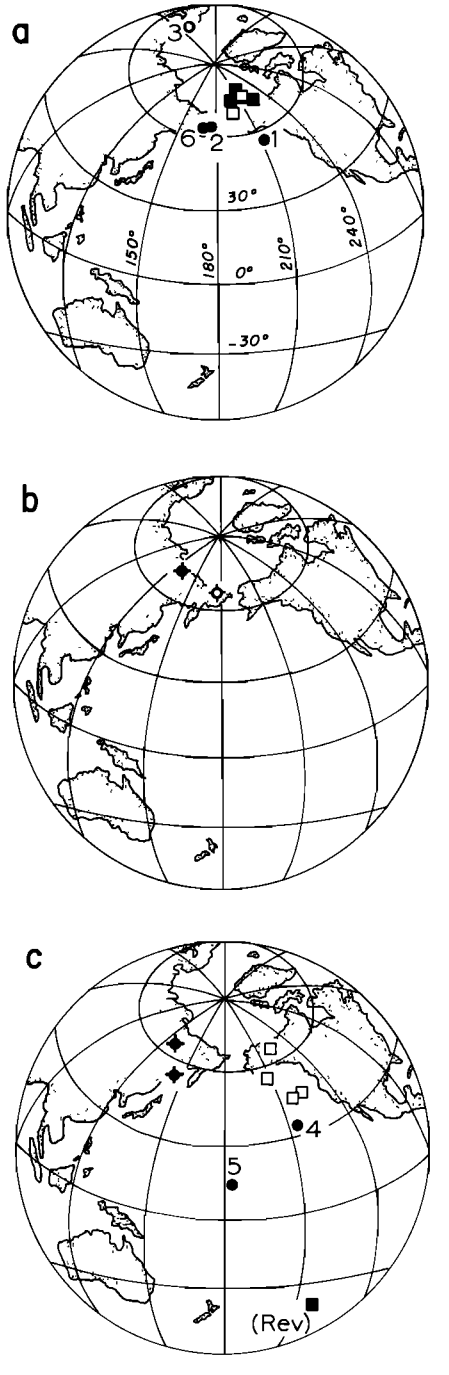


Fig. 14. (a) Middle to Late Jurassic paleopoles for the Korean Massifs (solid circles), the Okchon Folded Belt of Korea (open circle), the North China Block (solid squares), and the South China Block (open squares). The Korean paleopoles (with numbers 1, 2, 3 and 6) are from Table 2 and the China paleopoles are from Lin et al. [1985]. (b) Middle to Late Jurassic paleopoles for Eurasia (solid circle with cross) and North America transferred to Eurasia by closing the Atlantic Ocean (open circle with cross); the Eurasian pole is from Westphal et al. [1986: Jm-Ju in their Table 2], and the transferred Late Jurassic North American paleopole (66°N, 174°E) is from Van der Voo [1989a,b]. (c) Triassic paleopoles from the Ryongnam Massif of Korea (Table 2), from the North China Block, plotted as a southpole (Rev.) with respect to its usual location [Lin et al., 1985], the South China Block [Lin et al., 1985; Opdyke et al., 1986] and Eurasia [Westphal et al., 1986: Trl, Trm-u in their Table 6].

different, the Triassic paleopoles are significantly different (Figure 14c). In a similar vein, the Triassic paleopoles from South Korea and the SCB are somewhat different. These differences are primarily a function of the deviation in declination; the inclinations extrapolated for the same Korean reference site for all Triassic results are approximately the same. This implies that the differences in the Triassic paleopoles are chiefly to be explained by relative rotations, without any evidence that north-south movements occurred. The smallest rotations to explain the paleopole differences are clockwise with respect to the SCB and counterclockwise with respect to the NCB, but we note that the Triassic NCB paleopole plotted in Figure 14c is usually interpreted to be a south pole, i.e., of opposite polarity to what we infer for South Korea and the SCB. If this interpretation of the polarity is correct (and we have no reason to doubt it) then the post-Triassic rotation of South Korea with respect to the NCB is large (approximately 110°) and clockwise as well. Referring to the faunal correlations between the non-metamorphosed zone of the Okchon Folded Belt, the Pyongnam Basin of North Korea, and the North China Block, as mentioned above, we might note that while these rotations may be large,

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the paleolatitudes are all fairly similar; one could argue that the rotations are roughly about local vertical axes, so that their effects would not violate arguments in favor of reproductive communication between the Paleozoic faunal communities.

A previous paleomagnetic study, of the same Triassic rocks we studied, yielded west-northwesterly and upward directions [Shibuya et al., 1988]. Their paleopole, based on three sites, is also included in Figure 14c. However, the results do not demonstrate good clustering between sites and the applied tilt corrections do not improve the precision (Table 2). Because no details are presented in their paper, it is difficult to assess this problem, but it appears that the tilt corrections applied are questionable in view of the outcrop pattern in the Jangseong area. The results of Shibuya et al. [1988] show a higher inclination and, hence, paleolatitude, and show a declination which is ESE-WNW rather than ENE/WSW as in this study.

With respect to the Eurasian Triassic poles, the NCB, SCB and Korea all show significant deviations that are likely to be of tectonic origin; we conclude that the assembly of Asian elements was probably in full progress in the Triassic, but not completed until the Early Jurassic or later.

CONCLUSIONS

We conclude the following from our study:

1. The Late Jurassic paleomagnetic directions from the Okchon Belt deviate in a counterclockwise sense by about 60° from those of the adjacent Kyonggi and Ryongnam Massif, indicating a relative rotation. We interpret this rotation as due to left-lateral shear during the Daebo Orogeny.

2. Late Jurassic and Cretaceous paleopoles for Korea, the North China and South China blocks (with the exception of those from the Okchon Belt) are in rough agreement, indicating that no paleomagnetically detectable relative motions occurred after Late Jurassic time between these areas.

3. While there are significant differences between the Jurassic and Cretaceous paleopoles from Eurasia and those from Korea and the China Blocks, we agree with Courtillot and Besse [1986] that this may be due to the poor quality of the Eurasian reference poles.

4. Triassic paleopoles from South Korea, the North China Block, the South China Block and Eurasia are all different from each other. Most of these differences can be explained by local rotations of the blocks about vertical axes, without large north-south movements. Local rotations as well as relative north-south movements have been proposed for the China Blocks and Siberia on the basis of Permian paleopoles [Zhao and Coe, 1987]; apparently these motions were (partially) underway in the Triassic.

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Kwang Ho Kim, Department of Earth Sciences, Chonbuk National University, Chonju 560-756, South Korea.

Rob Van der Voo, Department of Geological Sciences, University of Michigan, Ann Arbor, MI 48109-1063.

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